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# TPE–604

## B. TECH. (PETROLEUM ENGINEERING)
## (SIXTH SEMESTER)
## MID SEMESTER EXAMINATION, April, 2023

PETROLEUM DRILLING ENGINEERING—II

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is directional drilling and deviation control ? What are the applications of directional drilling ? (CO1)

OR

(b) What are the parameters to be considered in directional well planning ? (CO1)

2. (a) What are the different types of well profile in directional drilling ? Explain with help of a neat diagram. (CO1)

OR

(b) It is desired to drill under the lake to the location designated for well 2. For this well build-and-hold trajectory will be used Horizontal departure to the target is 2655 ft at TVD of 9650 ft. The recommended rate of build is 2.0°/100 ft. The kickoff depth is 1600 ft. Determine ($b_1$) the radius of curvature $R_1$ ($b_2$) the maximum inclination angle $\theta$ ($b_3$) the measured depth to the end of build ($b_4$) the total measured depth.

(CO2)

3. (a) Derive the expression for the paraments to be required for geometrical planning of Type I well profile i.e build up and hold trajectory. (CO2)

OR

(b) Using the following information, calculate the coordinates of a Type I well profile : (CO2)

Slot coordinates 15.32 ft N, 5.06 ft E

Target coordinates 1650 ft N, 4510 ft E

TVD of target 9880 ft

KOP 1650 ft

Build up rate 1.5° per 100 ft

4. (a) What is positive displacement motors ? What are the advantages of using a motor with a bent housing rather than a bent sub, and for what applications might a bent housing be used ? (CO3)

OR

(b) What is whipstock ? What are its advantage and disadvantages ? Explain the pendulum principle of deviation control in directional drilling. (CO1)

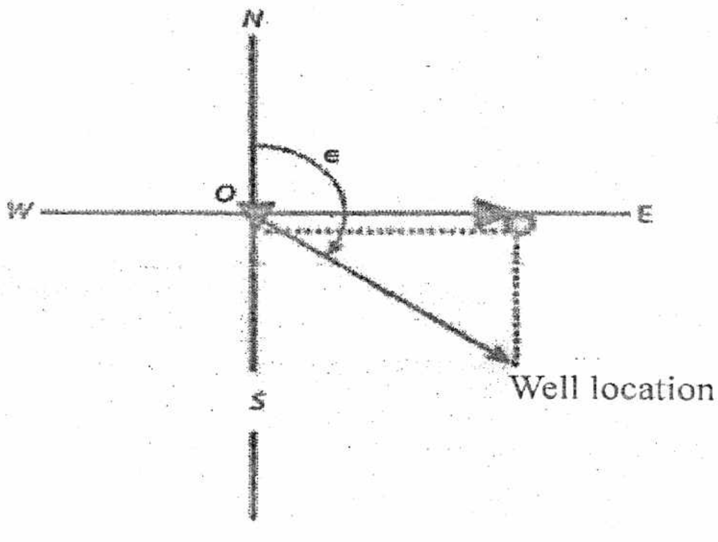
5. (a) What is bottom hole assembly (BHA) in directional drilling ? Discuss the three ways in which the BHA may be used for directional control.

(CO1)

OR

(b) A directional driller monitors the direction of a well from a reference location point O. The well has progressed 300 meters towards east and 500 meters towards south. What is the azimuth of the bottom of the well at this location ? What is the horizontal departure (closure distance) ?

(CO2)




Well direction

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# TPE–603

## B. TECH. (PETROLEUM ENGINEERING)
## (SIXTH SEMESTER)
## MID SEMESTER EXAMINATION, April, 2023

NATURAL GAS ENGINEERING

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the quality lines in phase diagram. Draw the P-T and P-V phase diagram for single component system showing the different region Define bubble and dew point in the reservoir of natural gas. (CO1)

OR

(b) Define the apparent molecular weight gas specific gravity, density and specific volume of the natural gas.

Solve the numerical :

A dry gas well is producing a gas steam of the following molar composition : 95 percent methane and 5 percent carbon dioxide. The molar mass of methane is 16 g/mol and that of carbon dioxide is 44 g/mol. Assuming Ideal gas behavior, gas constant R = 8.31 J/mol K. Find the gas stream density at $10^7$ Pa and 350 K in SI unit. (CO1)

2. (a) A gas well producing with the following composition : (CO2)

| Composition | Mole Fractions |
|---|---|
| C1 | 0.90 |
| C2 | 0.03 |
| C3 | 0.021 |
| $CO_2$ | 0.055 |

Assuming Ideal gas behavior, Calculate the following :

(i) Apparent molecular weight of the gas

(ii) Gas specific gravity

(iii) Gas density at 2000 psi and 150 degree F

(iv) Specific volume at 2000 psi and 150 degree F

OR

(b) Calculate the gas specific gravity of natural gas in the reservoir having the following composition : (CO2)

| Composition | Mole Fractions |
|---|---|
| C1 | 0.775 |
| C2 | 0.083 |
| C3 | 0.021 |
| $CO_2$ | 0.030 |
| $H_2S$ | 0.020 |

The reservoir is having at initial condition of Pi = 4000 psia and 190°F. By calculating sp. gr. of the gas also calculate Tpc, Ppc, Tpr, Ppr by using Standing and Katz correlations.

3. (a) What are the conventional natural gas production systems ? Write down the conditions in terms of pressure with respect to bubble point and dew point for the black oil, volatile oil, solution gas drive, wet gas and gas condensate. (CO2)

OR

(b) By which phenomena CBM is trapped in the bed seam ? Compare Coal Bed Methane (CBM) and Conventional reservoir on the different aspects. (CO2)

4. (a) How is gas hydrate formed ? Discuss the various issues related to extraction of natural gas hydrates. (CO1, CO2)

OR

(b) Gas hydrates extraction requires the unconventional approach, why ? Explain the different methods in the extraction of gas hydrate. (CO1, CO2)

5. (a) Explain hydraulic fracturing. What are the theory and techniques for the hydraulic fracturing ? Why hydraulic fracturing needed in the production of shale gas ? (CO2)

OR

(b) Discuss Dehydration of Natural Gas in detail. What are the effects of presence of water in natural gas ? (CO2)

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# MID SEMESTER EXAMINATION 2022-23

Name of the Course: B.Tech PE

Semeste VI[th]

Name of the Paper: DATA SCIENCE

Paper Code: TPE-602

Time: 1 Hour 30 Mins

Maximum Marks: 50

**Note: -**

(i) All questions are compulsory.
(ii) Attempt either (a) or (b)
(iii) Total marks in each main question are ten.
(iv) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10marks) | **CO1,CO2** |
| **(a)** | What are the types of Data Analytics? Explain each supported by the examples of industrial application. | |
| | **OR** | |
| **(b)** | Explain Normal Distribution in detail. Discuss random variable with an example. | |
| | | |
| **Q2** | (10marks) | |
| **(a)** | Explain Data Analysis process with neat diagram. What are Data Analysis methods? Define Qualitative and Quantitative methods. | **CO1, CO3, CO5** |
| | **OR** | |
| **(b)** | Explain correlation and describe the impact of outlier on correlation. | |
| | | |
| **Q3** | (10marks) | **CO1, CO3** |
| **(a)** | Explain null and alternative hypothesis by considering the example for a flipping coin. Explain rescaling of data with an example. | |
| | **OR** | |
| **(b)** | What are Data Analytics Techniques? What are main patterns in data while conducting Time Series Analysis? | |
| | | |
| **Q4** | (10 marks) | **CO1, CO3, CO6** |
| **(a)** | What is the significance of SQL in Data Science and Analytics? Explain the following SQL commands with examples: SELECT, INSERT, UPDATE, DELETE, CREATE | |
| | **OR** | |
| **(b)** | From the table name Customers do the following operations:<br>1.) Select Customer ID, Customer Name, Address and Postal Code | |

2.) Fetch the info (Customer ID, Contact Name, Postal Code) of the customers only from country India

| Customer ID | Customer Name | Contact Name | Address | City | Postal Code | Country |
|---|---|---|---|---|---|---|
| 1 | John Henkins | Adam Henkins | Obere St. 57 | Berlin | 1102 | Germany |
| 2 | Peter McGraw | Dillon McGraw | Avda. de la Constitución 2222 | Mexico | 3654 | Mexico |
| 3 | Dev Patil | Rahul Patil | Mataderos 2312 | Delhi | 21227 | India |
| 4 | Ahnaaf Hussain | Sareen Hussain | 120 Hanover Sq. | Hyderabad | 9567 | India |
| 5 | Yin Su | Quin Lee | Berguvsvägen 8 | Lulea | 21189 | Sweden |

| | | |
|---|---|---|
| **Q5** | (10 marks) | |
| **(a)** | What are Joins in SQL? Explain all with suitable examples for each. | **CO1, CO2, CO3, CO4, CO6** |
| | **OR** | |
| **(b)** | What are the challenges for Data Analytics in Oil & Gas industry? Write down some applications of Data Analytics in the similar domain. Give a use case scenario of Predictive modelling. | |
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# MID SEMESTER EXAMINATION 2022-23

Name of the Course: B.Tech PE

Semeste VI th

Name of the Paper: DATA SCIENCE

Paper Code: TPE-602

Time: 1 Hour 30 Mins

Maximum Marks: 50

**Note: -**

(i) All questions are compulsory.
(ii) Attempt either (a) or (b)
(iii) Total marks in each main question are ten.
(iv) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10marks) | **CO1,CO2** |
| **(a)** | What are the types of Data Analytics? Explain each supported by the examples of industrial application. | |
| | **OR** | |
| **(b)** | Explain Normal Distribution in detail. Discuss random variable with an example. | |
| | | |
| **Q2** | (10marks) | |
| **(a)** | Explain Data Analysis process with neat diagram. What are Data Analysis methods? Define Qualitative and Quantitative methods. | **CO1, CO3, CO5** |
| | **OR** | |
| **(b)** | Explain correlation and describe the impact of outlier on correlation. | |
| | | |
| **Q3** | (10marks) | **CO1, CO3** |
| **(a)** | Explain null and alternative hypothesis by considering the example for a flipping coin. Explain rescaling of data with an example. | |
| | **OR** | |
| **(b)** | What are Data Analytics Techniques? What are main patterns in data while conducting Time Series Analysis? | |
| | | |
| **Q4** | (10 marks) | **CO1, CO3, CO6** |
| **(a)** | What is the significance of SQL in Data Science and Analytics? Explain the following SQL commands with examples: SELECT, INSERT, UPDATE, DELETE, CREATE | |
| | **OR** | |
| **(b)** | From the table name Customers do the following operations:<br>1.) Select Customer ID, Customer Name, Address and Postal Code | |

| | | |
|---|---|---|
| | 2.) Fetch the info (Customer ID, Contact Name, Postal Code) of the customers only from country India | |

| Customer ID | Customer Name | Contact Name | Address | City | Postal Code | Country |
|---|---|---|---|---|---|---|
| 1 | John Henkins | Adam Henkins | Obere St. 57 | Berlin | 1102 | Germany |
| 2 | Peter McGraw | Dillon McGraw | Avda. de la Constitución 2222 | Mexico | 3654 | Mexico |
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| 4 | Ahnaaf Hussain | Sareen Hussain | 120 Hanover Sq. | Hyderabad | 9567 | India |
| 5 | Yin Su | Quin Lee | Berguvs vägen 8 | Lulea | 21189 | Sweden |

| | | |
|---|---|---|
| **Q5** | (10 marks) | |
| **(a)** | What are Joins in SQL? Explain all with suitable examples for each. | **CO1, CO2, CO3, CO4, CO6** |
| | **OR** | |
| **(b)** | What are the challenges for Data Analytics in Oil & Gas industry? Write down some applications of Data Analytics in the similar domain. Give a use case scenario of Predictive modelling. | |
| | | |

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# TPE–601

## B. TECH. (PETROLEUM ENGINEERING)

## (SIXTH SEMESTER)

## MID SEMESTER EXAMINATION, April, 2023

RESERVOIR MODELLING AND SIMULATION

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the three essential properties of a reservoir rocks ? Explain each in detail. Name the various type of physical properties of rock.

(CO1)

OR

(b) What are the different types of sedimentary rocks are available on the earth ? Explain their composition and relative abundance on the earth.

(CO1)

2. (a) Explain a petroleum system. Draw a cross section of a petroleum system and explain its all important components. (CO1)

*P. T. O.*

OR

(b) Draw a diagram showing Generation, Migration and trapping of hydrocarbons of a Petroleum system and explain its each components in detail. (CO1)

3. (a) What is the difference between classic reservoir engineering, reservoir modeling and Reservoir simulation ? Why are all required ? (CO2)

OR

(b) Name various drive mechanisms of petroleum system. Explain Primary and secondary recovery mechanism. (CO2)

4. (a) Define with figure cubical packing, orthorhombic packing and rhombohedra-packing of spheres. How these packing effect porosity and permeability ? (CO2)

OR

(b) Explain Porosity, Total and Effective Porosity, Permeability, and Oil saturation. (CO2)

5. (a) Draw a sequential reservoir modeling workflows for building a reservoir model. Explain its each component. (CO2)

OR

(b) Derive single phase linear flow equation in a porous cylindrical rock. Explain and draw a plot of water saturation profile in an oil-water system. (CO2)

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# TPE–611

## B. TECH. (PETROLEUM ENGG.) (SIXTH SEMESTER)

## MID SEMESTER EXAMINATION, April, 2023

### ALTERNATE ENERGY RESOURCES

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Enumerate the criteria based on which energy can be classified. Give their availability, relative merits, demerits and their classification. (CO1/CO2/CO3)

OR

(b) What is meant by renewable energy sources ? What are the prospects of alternate energy sources in India ? (CO1/CO2/CO3)

2. (a) Distinguish between renewable and non-renewable energy sources. (CO3/CO4)

OR

(b) (i) Define beam, diffuse and global radiation.

(ii) Explain the depletion process of solar radiation as it passes through the atmosphere to reach at the surface of the earth. (CO3/CO4)

*P. T. O.*

3. (a) Compare basic features among "Pyranometer, Pyrheliometer and Sunshine recorder". (CO1/CO4)

OR

(b) Classify different types of solar thermal collectors and show the constructional details of a fiat plate collector. What are its main advantages ? (CO1/CO4)

4. (a) Describe the principle of solar photovoltaic energy conversion with a neat diagram. (CO4)

OR

(b) (i) Describe the classification of solar cells based on the type of active material used.

(ii) How are electron-hole pairs generated when solar radiation is directed on a semiconductor material ? (CO4)

5. (a) What are major advantages and disadvantages of a solar PV system ? (CO4)

OR

(b) Draw a schematic diagram of solar pond based electric power plant and explain its working. (CO4)

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# TPE–605

## B. TECH. (PETROLEUM ENGINEERING)
## (SIXTH SEMESTER)
## MID SEMESTER EXAMINATION, April, 2023

PETROLEUM PRODUCTION ENGINEERING–II

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) In deep water with depth of 4500 ft in offshore, which platform should be used ? Describe that platform and give reason for your answer. (CO1)

OR

(b) In deep water with depth of 510 m in offshore, which platform should be used ? Describe that platform and give reason for your answer. (CO1)

2. (a) Explain what is well activation and how compressed air can be used for well activation. (CO2)

OR

(b) Describe subsea production system, field architecture and subsea wellheads. (CO2)

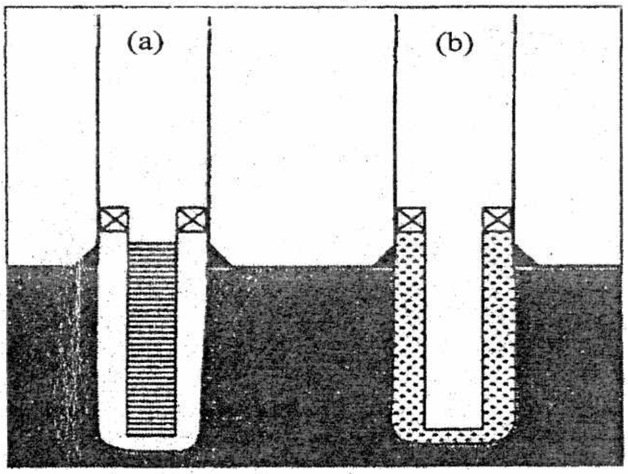
3. (a) Differentiate between SCR and TTR. (CO1)

OR

(b) Why there is need of an FPSO in offshore ? Also describe different modules of an FPSO. (CO1)

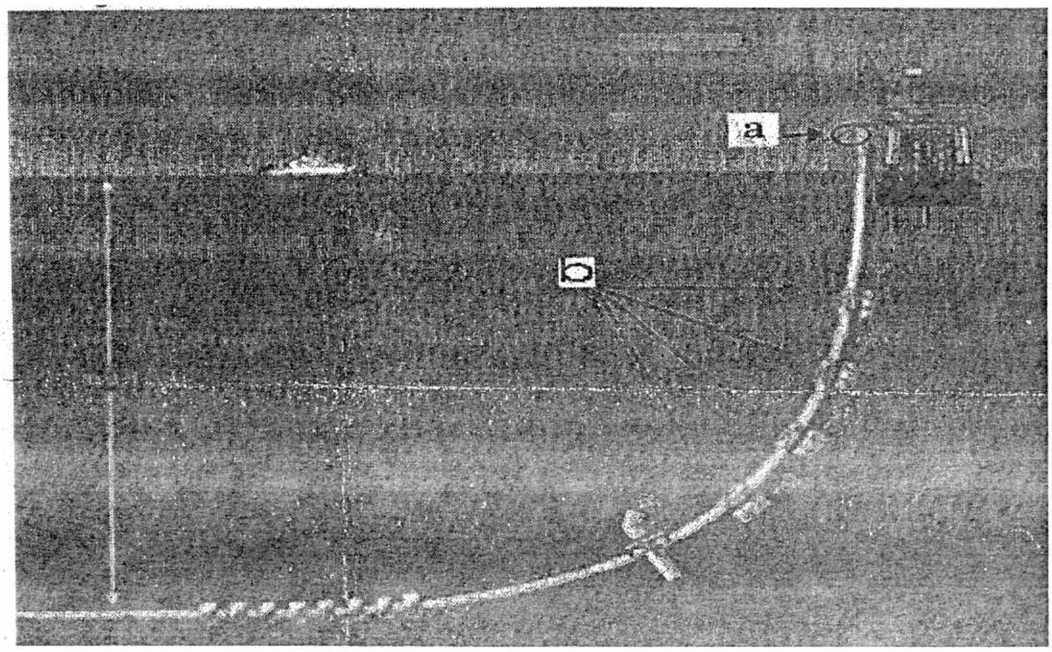
4. (a) From the given figure identify the types of liner and differentiate between them.

(CO2)




OR

(b) From the given figure identify and explain type of riser and also identify a & b in the given figure. Also write why it is used in riser. (CO2)



5. (a) In deep water with depth upto 2300 m in offshore, which platform should be used ? Describe that platform and give reason for your answer. (CO1, CO3)

OR

(b) Explain well deliverability and forecast of well production. (CO1, CO3)